Thread the Needle: Cues-Driven Multiassociation for Remote Sensing Cross-Modal Retrieval

Yaxiong Chen, Jirui Huang, Zhaoyang Sun, *Graduate Student Member, IEEE*, Shengwu Xiong, and Xiaoqiang Lu, *Senior Member, IEEE*

***Abstract*— Rapid advances in Earth observation technologies have yielded numerous remotely sensed images and corresponding text data, enabling cross-modal image–text retrieval to extract valuable clues. However, current methods often focus on learning global semantic information from text and remote sensing (RS) images, while neglecting fine-grained semantic alignment and correlation. In addition, contrastive learning between modalities is often insufficient. To address these issues, we propose an innovative cues-driven multiassociation feature matching network (CDMAN) for cross-modal RS image retrieval. The proposed method primarily involves two key steps: 1) aligning positive samples and enhancing fusion for negative samples based on modal cues. To achieve precise alignment between RS images and text and facilitate the learning process for negative samples in contrastive learning, we have developed a novel fine-grained cues injection module that aligns and guides modalities using fine-grained cues; and 2) establishing multigranularity associative learning. To address the issue of insufficient association between RS images and text, we have implemented multigranularity collaborative associative learning, focusing on general and fine-grained modal associations. By fully leveraging modal cues, our method maintains both detailed associations and overall consistency in global associations. Experiments demonstrate that, compared to baseline methods, this approach achieves more accurate cross-modal retrieval (MCR) by combining fine-grained alignment and multigranularity associations.**

***Index Terms*— Cross-modal image–text retrieval, deep features similarity, local bidirectional alignment, local–global association learning.**

Received 7 May 2024; revised 3 August 2024 and 20 October 2024; accepted 9 November 2024. Date of publication 2 December 2024; date of current version 10 December 2024. This work was supported in part by NSFC under Grant 62101393 and Grant 62176194; in part by the National Key Research and Development Program of China under Grant 2022ZD0160604; in part by the Key Research and Development Program of Hubei Province under Grant 2023BAB083; in part by the Project of Sanya Yazhou Bay Science and Technology City under Grant SCKJ-JYRC-2022-76 and Grant SKJC-2022-PTDX-031; in part by the Project of Sanya Science and Education Innovation Park of Wuhan University of Technology under Grant 2021KF0031 and Grant 2022KF0020; in part by Hainan Province "Nanhai New Star" Technology Innovation Talent Platform Project under Grant NHXXRCXM202361; in part by the Youth Fund Project of Hainan Natural Science Foundation under Grant 6220N344; in part by the Natural Science Foundation of Chongqing under Grant cstc2021jcyj-msxmX1148; in part by the MindSpore, which is a New Deep Learning Computing Framework (https://www.mindspore.cn/); in part by the National Natural Science Foundation of China under Grant 62271484; in part by the National Science Fund for Distinguished Young Scholars under Grant 61925112; in part by the Key Research and Development Program of Shaanxi under Grant 2023-YBGY225; and in part by the High Performance Computing Platform of YZBSTCACC. *(Corresponding author: Shengwu Xiong.)*

Yaxiong Chen is with Sanya Science and Education Innovation Park, Wuhan University of Technology, Sanya 572000, China, also with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan 430070, China, and also with the Interdisciplinary Artificial Intelligence Research Institute, Wuhan College, Wuhan 430212, China.

Jirui Huang and Zhaoyang Sun are with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan 430070, China, also with Sanya Science and Education Innovation Park, Wuhan University of Technology, Sanya 572000, China, and also with Chongqing Research Institute, Wuhan University of Technology, Chongqing 401122, China.

Shengwu Xiong is with the Interdisciplinary Artificial Intelligence Research Institute, Wuhan College, Wuhan 430212, China, also with Shanghai Artificial Intelligence Laboratory, Shanghai 200232, China, and also with the School of Information Science and Technology, Qiongtai Normal University, Haikou 571127, China (e-mail: xiongsw@whut.edu.cn).

Xiaoqiang Lu is with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China.

Digital Object Identifier 10.1109/TGRS.2024.3509639

I. Introduction

With increasing volume of remote sensing (RS) data, efficient data analysis technology has become a critical focus for researchers [1], [2], [3], underscoring the importance of extracting valuable knowledge from extensive RS datasets [4], [5], [6]. The cross-modal retrieval models (CMRMs) aim to retrieve data from one modality (e.g., text, images, and audio) using another modality, enabling the association and translation of information across different modalities. Its significance lies in meeting the growing demand for multimodal data integration and application, driving advancements in technologies such as search engines, recommendation systems, and multimodal generative models. However, directly applying existing CMRMs to RS data presents several specific challenges. Unlike natural image data, RS images have distinct characteristics, such as containing extremely small objects (sometimes as few as 10 pixels) and requiring sophisticated spatial reasoning due to their top-down view. Consequently, employing natural image-trained CMRMs for RS text–image retrieval encounters substantial difficulties. Additionally, achieving precise alignment between different modalities and establishing accurate cross-modal relationships remain major challenges in RS CMRMs.

In recent years, numerous CMRMs for RS have demonstrated promising results. Current cross-modal RS image retrieval approaches can be categorized by their feature extraction techniques and feature interaction mechanisms. Specifically, based on the type of feature extractor, they are divided into deep convolutional neural network (DCNN)-based methods and Transformer-based methods. DCNN-based

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:53:06 UTC from IEEE Xplore. Restrictions apply.

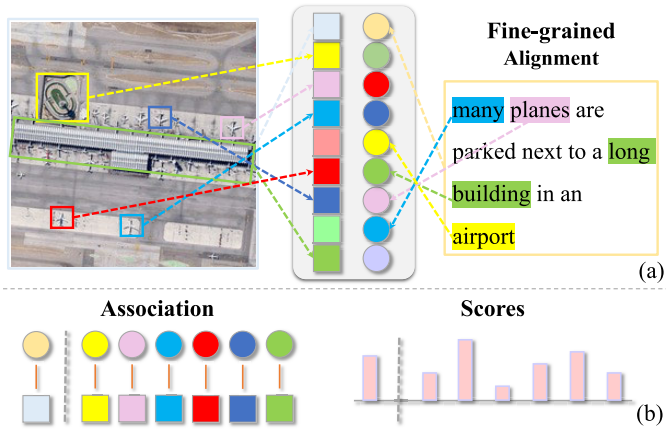


Fig. 1. (a) Fine-grained alignment of cross-modal features. (b) Multigranularity association by calculating the score between local and global RS images and text from the same semantic center.

approaches, such as the asymmetric multimodal feature matching network (AMFMN) [7] and the global and local information (GaLR) model [8], typically utilize DCNNs to extract feature representations from RS images, while recurrent neural networks (RNNs) are employed to extract text features [9]. However, due to the inductive bias [10] inherent in convolutional networks, the aforementioned methods struggle with flexible global context modeling, limiting their ability to effectively capture the overall information in RS images. Conversely, Transformer-based methods, such as the interacting–enhancing feature Transformer (IEFT) [11], utilize Transformer architectures to extract feature representations for both images and texts, offering a unified framework for multimodal data processing.

Cross-modal RS image–text retrieval methods can also be divided based on the type of feature interaction into global interaction-based methods and local interaction-based methods. Global interaction-based methods, such as [12] and [13], achieve interaction by first extracting global features and then performing global alignment and association. While these methods exhibit better overall consistency, the regions of interest in RS images often occupy only a small portion of the pixels, making it challenging for traditional global alignment models to capture complex semantic relationships effectively. In contrast, local interaction-based methods, such as the multiscale interactive Transformer (MSIT) model [14], learn multiscale features of RS images to uncover potential relationships between different modalities, achieving local alignment. Local alignment not only adds to the computational cost, but also, due to the inherent asymmetry between images and text, often results in aligning multiple local image regions with textual features in a one-to-one manner. However, since not all local regions are strongly correlated with the text, this approach may introduce unnecessary noise. In the final stage, these methods typically employ global association for interaction. Yuan et al. [15], addressing computational efficiency concerns, designed a lightweight cross-modal retrieval (LW-MCR) model. While this model introduces a novel hidden supervision optimization through knowledge distillation, it still overlooks the computational costs associated with local alignment. In later work, Cheng et al. [16] developed a semantic alignment module and applied attention and gating mechanisms to selectively enhance data features, aiming to achieve more discriminative feature representations. Lv et al. [17] recognized the complementary nature of information between modalities and proposed a new fusion-based correlation learning model (FCLM). Although these approaches have made progress in feature representation and cross-modal information integration, they do not account for the fact that not all local regions in RS images are closely related to textual descriptions. For instance, as shown in Fig. 1(a), neither the objects indicated by the yellow rectangular box nor the large background area of the airport runway are mentioned in the text. Moreover, due to the minimal interclass variability in RS images, the buildings highlighted by the yellow box are often misclassified as "airports." Therefore, when constructing positive and negative image–text pairs in contrastive learning, special emphasis should be placed on adequately learning from negative pairs.

Traditional global alignment methods frequently struggle to capture intricate associations, while local alignment can introduce noise and asymmetry issues. To make up for these deficiencies, a solution that synergizes both global and local associations is essential, particularly for capturing the unique multiscale features and subtle semantic variations inherent in RS data. Given the subtle differences among ground object categories, constructing challenging negative samples is essential for improving model robustness. Thus, we focus on leveraging cues between text and RS images to achieve precise alignment and association between query text (or RS images) and their corresponding counterparts [18], as shown in Fig. 1(b).

In our study, we propose the cues-driven multiassociation feature matching network (CDMAN), which aims to facilitate efficient and accurate cross-modal RS image retrieval through multigranularity intermodal alignment and association, as illustrated in Fig. 2. The CDMAN captures correlations between RS images and texts at multiple granularity levels while emphasizing the significance of negative samples in contrastive learning to enhance the model's comprehension of complex data. First, we have designed a distinctive cues injection module to promote learning between extracted global and local features. This module utilizes detailed modal information as cues to achieve fine-grained alignment among positive sample pairs. Additionally, to bolster the model's capacity for recognizing negative samples, we constructed challenging negative samples that underscore RS images with minimal interclass differences, yet significant intraclass differences, ensuring effective learning from these complex cases. Additionally, we have implemented a multigranularity collaborative association mechanism to enhance intermodal connections during retrieval, thereby ensuring the effective capture of both global features and critical details. This approach addresses the limitations associated with missing detailed associations in the global alignment process.

Overall, our contribution can be summarized as follows.

1) We present the CDMAN for cross-modal RS image retrieval, which achieves intermodal alignment and association at multiple levels of granularity while enhancing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:53:06 UTC from IEEE Xplore. Restrictions apply.

attention to and learning from negative samples in the contrastive learning process.
2) We introduce a novel fine-grained cues injection module that leverages detailed modal information to achieve precise alignment of positive sample pairs across modalities and construct challenging negative samples.
3) We develop a multigranularity cooperative association mechanism that enhances modal associations by leveraging both coarse- and fine-grained associations.
4) Experimental results on the RSITMD, RSICD, and UCM image–text datasets demonstrate that our method achieves better retrieval precision than other state-of-the-art approaches.

The remainder of the article is divided into three sections. Section II gives an elaboration of the detailed proceeding of the proposed CDMAN method. Section III elaborates on the details of the experiments and the experimental results. Section IV shows the conclusion for the CDMAN method.

II. Proposed Method

A. Problem Definition

Given N pairwise units, $\mathcal{P} = \{I_n, T_n\}_{n=1}^{N}$, where I_n represents the nth RS image instance and T_n represents the nth text instance. The purpose of the proposed CDMAN approach is to identify a suitable projection for RS image modality and text modality while preserving the similarity of cross-modal samples [19], [20]. Specifically, the higher the similarity between the nth RS image instance and the nth text instance, the greater the similarity score between I_n and T_n.

B. RS Image Feature Extraction

Text descriptions often mention multiple targets in RS images, which are widely distributed. Existing cross-modal RS image–text retrieval methods relying on convolution operations face limited receptive fields, making it challenging to capture long-range object information. In contrast, the proposed RS image feature extraction module effectively captures long-range information and learns contextual semantic features, as illustrated in Fig. 2. The RS image feature extraction architecture comprises four modules, each with a similar network structure containing a patch embedding layer and Transformer encoder layers. In each module, the input is divided into m patches, which are flattened and passed through the patch embedding layer. The embedded patches are then processed by a Transformer encoder, and the output is reshaped into feature maps representing each module's output. The patch embedding layer serializes the input, preserving local feature continuity across stages. It includes a linear projection and a layer normalization (Norm) operation [21]. The detailed operation of the Transformer encoder layers is shown in Fig. 2. First, the output P of the patch embedding layer is linearly mapped into a query Q_1. To reduce the computational overhead of the attention operation, the output P undergoes a convolution layer with a layer normalization operation and a linear projection to obtain a key K_1 and a value V_1. The entire process can be expressed as follows:

$$K_1 = \rho(\mathcal{N}(\mathrm{Cov}_k(P, R_1, R_2))) \tag{1}$$

$$V_1 = \rho(\mathcal{N}(\mathrm{Cov}_v(P, R_1, R_2))) \tag{2}$$

where $\mathcal{N}(\cdot)$ denotes the layer normalization operation [22]. $\rho(\cdot)$ denotes the linear projection operation. $\mathrm{Cov}_k(\cdot)$ represents the convolution operation for K_1. $\mathrm{Cov}_v(\cdot)$ represents the convolution operation for V_1. R_1 denotes the size of the convolution kernel. R_2 denotes the stride of the convolution.

Second, Q_1, K_1, and V_1 are input into the multihead attention (MHA) for further encoding. Details of the MHA can be formulated as follows:

$$\mathrm{MHA}(Q_1, K_1, V_1) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_H)W^o \tag{3}$$

$$\mathrm{head}_h = \mathrm{Attention}\left(Q_1 W^Q, K_1 W^K, V_1 W^V\right) \tag{4}$$

where $\mathrm{Concat}(\cdot)$ denotes the concatenation operation [23]. W^o, W^Q, W^K, and W^V represent linear projection parameters. H denotes the number of heads in the attention layer.

Finally, the add operation is applied to the output P of the patch embedding layer and $\mathrm{MHA}(Q_1, K_1, V_1)$, followed by layer normalization, a feedforward neural network, and another add operation. Subsequently, the output of the Transformer encoder layers is expressed as follows:

$$O = \mathrm{FF}(\mathcal{N}(\mathrm{MHA}(Q_1, K_1, V_1) + P)) + (\mathrm{MHA}(Q_1, K_1, V_1) + P) \tag{5}$$

where O denotes the output of the Transformer encoder layers. $\mathrm{FF}(\cdot)$ refers to a fully connected layer known as a feedforward neural network.

In summary, the input RS image is processed through four modules. The output of the fourth module is the local features of the RS image, $F_l(I_n)$. The flattened local features $F_l(I_n)$ are then fed into a linear projection to obtain the global features of the RS image, $F_g(I_n)$. On this basis, the Transformer encoder layers focus on multiple objects through an MHA mechanism to extract contextual semantic feature representations.

C. Text Feature Extraction

The text feature extraction branch of the deep cross-modal RS image–text retrieval framework is illustrated in Fig. 2. Following [24], input sentences are processed through a tokenization operation. Additionally, two special tokens are added at the start ([CLS]) and the end ([SEP]) of the tokenized result. The [CLS] token denotes the meaning of an entire sentence and can represent the global features of the text. Token embeddings, segment embeddings, and position embeddings are summed for each token to generate its input representation, serving as the BERT model's input. The BERT model architecture consists of a multilayer bidirectional Transformer encoder. In the BERT model, the number of layers (i.e., Transformer blocks) is B, the hidden size is H, and the number of self-attention heads is A. The token representations from the BERT model are fed into a linear projection to obtain textual local features $F_l(T_n)$. The [CLS] representation from the BERT model is fed into an output layer to obtain textual global features $F_g(T_n)$.

D. Fine-Grained Cues Injection

To maximize the similarity between positive sample pairs and avoid the problem of too many simple samples [19],

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:53:06 UTC from IEEE Xplore. Restrictions apply.

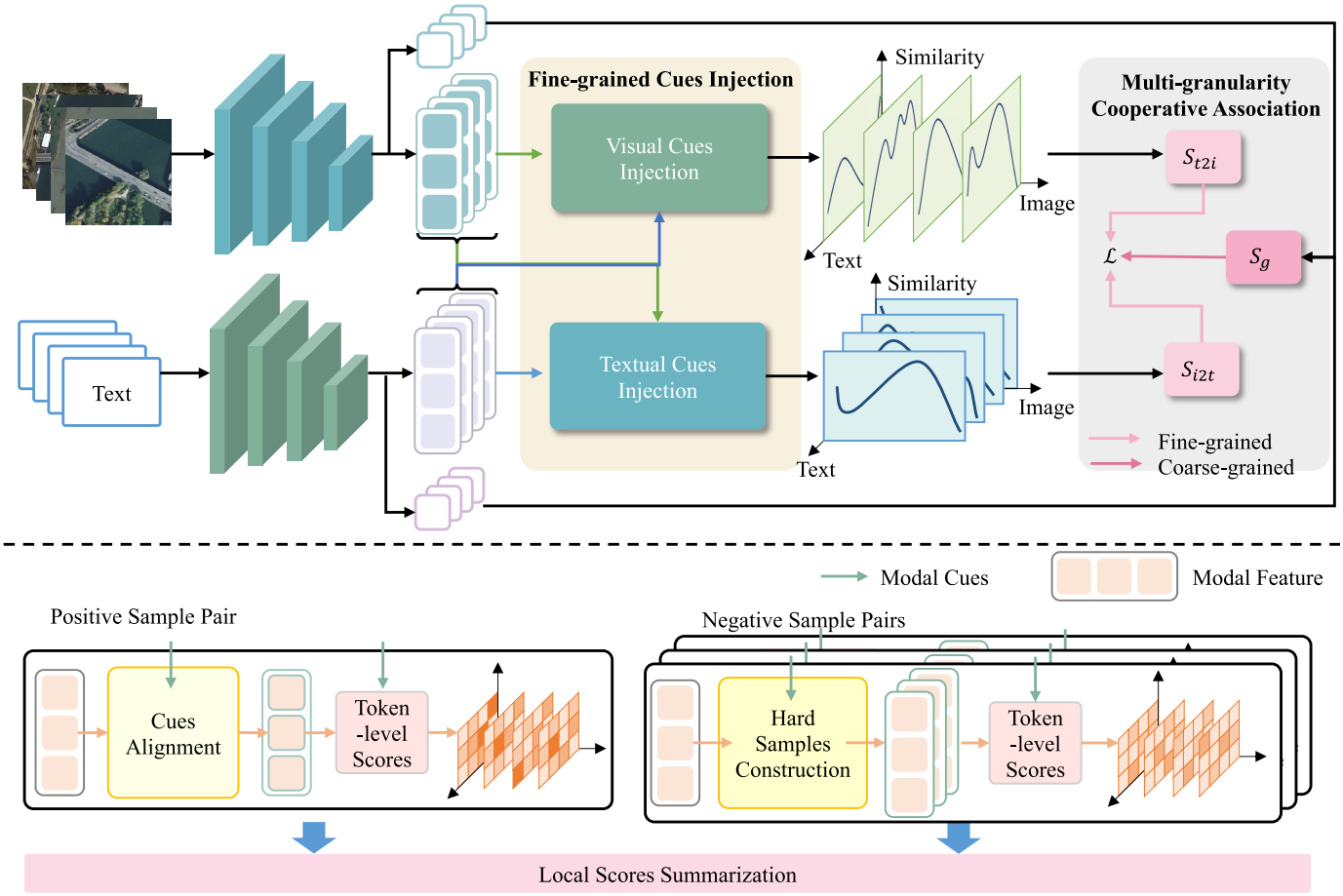


Fig. 2. Overview of the CDMAN framework. The CDMAN framework accepts RS image–text pairs as input. The framework consists of an RS image feature extraction module, text feature extraction module, fine-grained cues injection module, and multigranularity cooperative association module. The proposed CDMAN framework can learn contextual semantic feature representation and perform fine-grained alignment and multigranularity association between RS images and texts.

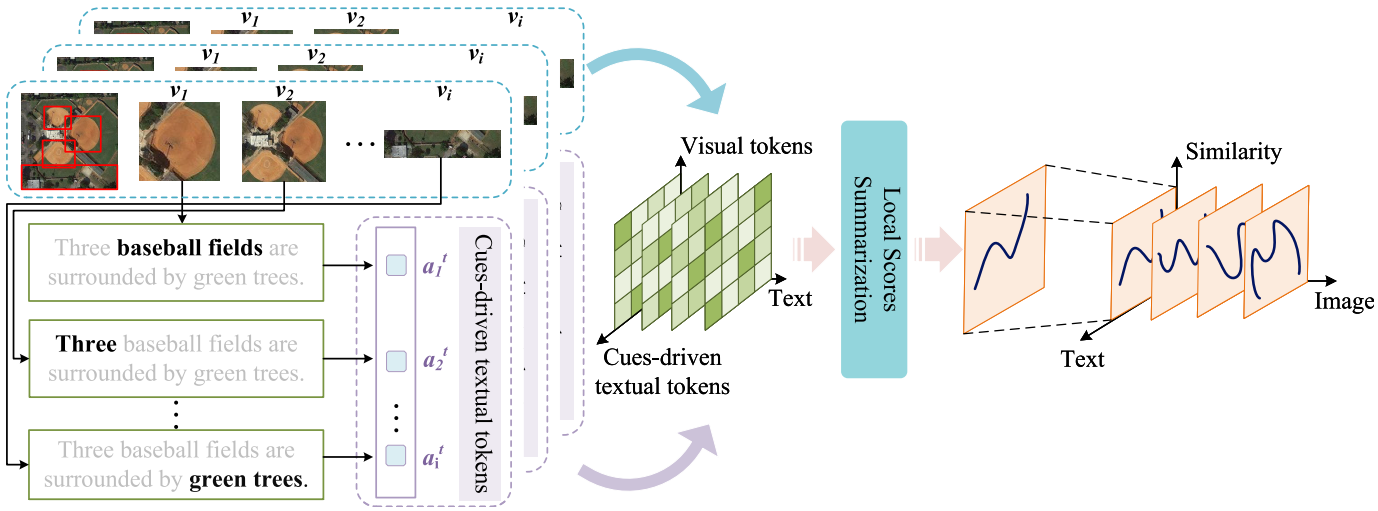


Fig. 3. Visual cues injection. Utilize fine-grained image cues to achieve alignment with positive text samples, while constructing difficult samples for negative text samples.

[25], we introduce a fine-grained cues injection stage, wherein visual cues and text injections are performed, as shown in Fig. 2. For positive sample pairs, cues injection aids in information alignment between modalities. For negative samples, cues injection transforms them into difficult negative samples, upon which comparative loss constraints are applied, thereby enhancing the model's ability to distinguish between negative samples [26], [27].

1) Visual Cues Injection: The rich semantic information in RS images cannot be effectively utilized in global alignment;

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:53:06 UTC from IEEE Xplore. Restrictions apply.

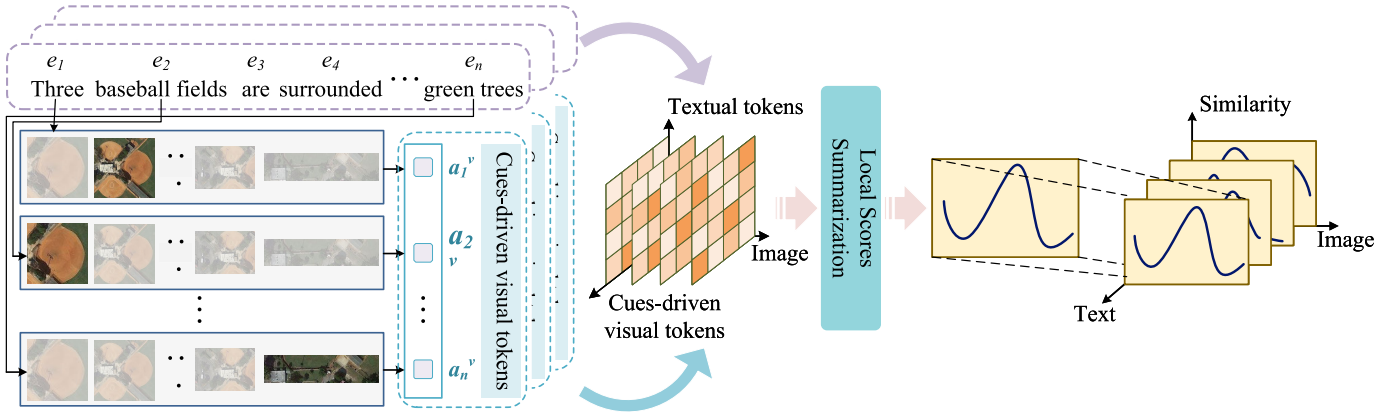


Fig. 4. Textual cues injection. Utilize fine-grained text cues to achieve alignment with positive image samples, while constructing difficult samples for negative image samples.

therefore, we have developed a visual cues injection module. This module aids positive sample pairs in perceiving detailed information from RS images, thereby facilitating semantic alignment between modalities. Additionally, unpaired RS images are employed to increase the difficulty of discriminating text-negative samples, thereby enhancing the effectiveness of contrastive learning, as shown in Fig. 3. RS image cues features, $F_l(I_n)$, contain a set of features from each patch of an image, where $F_l(I_n) = i_1, i_2, \ldots, i_M$. Text local features, $F_w(T_n)$, contain a set of word features from each word of a sentence, where $F_w(T_n) = t_1, t_2, \ldots, t_K$. The visual cues module includes three stages: visual cues alignment for positive samples, hard text sample construction for negative samples, and local scores summarization. The specific process is described as follows.

a) Visual cues alignment for positive samples: This process is illustrated in Fig. 3. Step 1 focuses on identifying words related to each image patch. Step 2 evaluates the relevance of each image patch to the corresponding attention statement vector. Cosine similarity is used to measure the similarity between local features of RS images and text. The procedure is as follows:

$$s_{mk} = \frac{\tau(i_m)t_k}{\|i_m\|\|t_k\|} \tag{6}$$

where s_{mk} denotes the similarity between the mth cue and the kth word. τ denotes the transpose operation. Inspired by [28], normalizing the similarity matrix is beneficial. The process is formulated as follows:

$$\hat{s}_{mk} = \frac{\beta(s_{mk})}{\sqrt{\sum_{m=1}^{M}(\beta(s_{mk}))^2}} \tag{7}$$

where β denotes the max operation and $\hat{s}_{mk}$ represents the normalized similarity of visual cues alignment for positive samples.

To obtain the aligned word features [29], we use the normalized similarity of visual cues alignment to filter and weight the words of interest in the text, and ultimately obtain the cues-driven textual tokens embedding

$$f_m^t = \alpha_{mk} t_k \tag{8}$$

$$\alpha_{mk} = \frac{\exp(\gamma_1 \hat{s}_{mk})}{\sum_{k=1}^{K}\exp(\gamma_1 \hat{s}_{mk})} \tag{9}$$

where f_m^t denotes the mth cues-driven textual token embedding. γ_1 denotes the inverse temperature for the softmax function [30].

Most existing methods directly perform global association or implicit local association, which may lead to insufficient association between modalities. Therefore, to maintain explicit fine-grained association, we utilize the available visual cues and text tokens to compute their similarity scores. This process is described as follows:

$$R(i_m, f_m^t) = \frac{\tau(i_m) f_m^t}{\|i_m\|\|f_m^t\|} \tag{10}$$

where $R(i_m, f_m^t)$ denotes the relevance between the mth region and the sentence.

b) Hard text samples construction for negative samples: Unlike the existing methods of positive and negative sample augmentation, we focus on hard sample mining. However, directly mining hard samples can lead to instability in the training process and even overfitting. Therefore, we utilize existing data to construct hard samples for text with the help of visual cues. The specific construction process is similar to the alignment flow of positive samples with visual cues. Each visual cue calculates its similarity and weights with all local text features in the same batch, thereby obtaining the similarity score between each cues-driven text token and the visual cues.

c) Local scores summarization: Inspired by [31], the final similarity score is computed using Log-SumExp pooling [32]. The process is formulated as follows:

$$S_{TI} = \log\left(\sum_{m=1}^{M}\exp(\gamma_2 R(i_m, f_m^t))\right)^{\frac{1}{\gamma_2}} \tag{11}$$

where S_{TI} represents the final similarity score for visual cues injection and γ_2 denotes the margin parameter.

2) Textual Cues Injection: To address challenges in RS images, such as occlusion and illumination changes, we propose a textual cues injection module. This module incorporates textual information into visual processing, enhancing the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:53:06 UTC from IEEE Xplore. Restrictions apply.

model's ability to accurately interpret local details and achieve fine-grained semantic alignment. Similar to visual cues injection, the textual cues injection module comprises three phases: textual cues alignment for positive samples, hard image sample construction for negative samples, and local scores summarization. The process is described below.

a) Textual cues alignment for positive samples: This formulation is illustrated in Fig. 4. First, we focus on the RS image regions related to each word to achieve the injection of textual cues into the image. To carry out the process of textual cues injection, similarity normalization is defined as follows:

$$\widetilde{s}_{mk} = \frac{\beta(s_{mk})}{\sqrt{\sum_{k=1}^{K}(\beta(s_{mk}))^2}} \tag{12}$$

where $\widetilde{s}_{mk}$ represents the normalized similarity for textual cues alignment for positive samples and β denotes the max operation.

To focus on RS image regions [29], the weighted combination of image region representations is defined as follows:

$$f_k^i = \sum_{m=1}^{M} \widetilde{\alpha}_{mk} i_m \tag{13}$$

$$\widetilde{\alpha}_{mk} = \frac{\exp\left(\gamma_1 \widetilde{s}_{mk}\right)}{\sum_{m=1}^{M} \exp\left(\gamma_1 \widetilde{s}_{mk}\right)} \tag{14}$$

where f_k^i denotes the cues-driven visual token embedding.

To preserve the explicit guidance of textual cues for RS images, the cosine similarity function is used to calculate the similarity between the cues-driven visual tokens f_k^i and the textual cues t_k. The process is formulated as follows:

$$R\left(t_k, f_k^i\right) = \frac{\tau(t_k) f_k^i}{\|t_k\| \left\| f_k^i \right\|} \tag{15}$$

where $R(t_k, f_k^i)$ denotes the relevance between the kth word and the RS image.

b) Hard image samples construction for negative samples: Consistent with the construction process of hard text samples for negative samples, each textual cue undergoes similarity calculation and weighting with all the local image features in the same batch. This results in obtaining the similarity scores between each cues-driven visual token and the corresponding textual cues.

c) Local scores summarization: Inspired by [33], the final similarity score is computed using average pooling. The process is formulated as follows:

$$S_{IT} = \frac{\sum_{k=1}^{K} R\left(t_k, f_k^i\right)}{K} \tag{16}$$

where S_{IT} denotes the final similarity scores for textual cues injection.

E. Multigranularity Cooperative Association

To address the limitations of single-granularity association in handling intermodal relationships, we introduce a multigranularity cooperative association mechanism. This mechanism considers information from multiple levels simultaneously, including overall image features, local image features, textual vocabulary, syntax, and semantics, while also better addressing the heterogeneity issue between different modalities. Inspired by [34], coarse-grained association learning can be performed using the cosine similarity function for RS image global features $F_g(I_n)$ and text global features $F_g(T_n)$. Cues-driven fine-grained association learning can be performed through fine-grained cues injection for RS image local features $F_l(I_n)$ and text local features $F_l(T_n)$. The formulation of the process is expressed as follows:

$$S(I, T) = \omega S_g + \rho(S_{TI} + S_{IT}) \tag{17}$$

$$S_g = \frac{F_g(I_n) F_g(T_n)}{\left\|F_g(I_n)\right\| \left\|F_g(T_n)\right\|} \tag{18}$$

where $S(I, T)$ denotes the multigranularity cooperative similarity between RS images and texts. S_g denotes the coarse-grained similarity score between $F_g(I_n)$ and $F_g(T_n)$. ω represents a factor governing the influence of the coarse-grained similarity score. ρ represents a parameter determining the influence of the fine-grained similarity score.

The objective of the CDMAN approach we put forward is to identify an appropriate projection function for the image modality and the text modality while maintaining the similarity of the cross-modal sample [35]. To achieve this objective, the similarity between positive RS image–text pairs should be maximized, and the proximity of negative RS image–text pairs should be minimized [36]. Inspired by [37], a bidirectional triplet objective function based on reconstructed hard samples is developed to maximize the similarity of positive RS image–text pairs and minimize the distance of negative pairs. The objective function is given as follows:

$$L(I, T) = \sum_{\hat{T}} \left[\mu - S(I, T) + S\left(I, \hat{T}\right)\right]_+ + \sum_{\hat{I}} \left[\mu - S(I, T) + S\left(\hat{I}, T\right)\right]_+ \tag{19}$$

where μ denotes the margin threshold parameter. The first sum considers all negative sentences $\hat{T}$ for an RS image I. The second sum considers all negative RS images $\hat{I}$ for a sentence T. Here, $[x]_+ = \max(x, 0)$. The proposed CDMAN algorithm is outlined in Algorithm 1. The total objective function $L(I, T)$ is optimized by the Adam optimizer [38]. Algorithm 1 details the steps of the optimization process for the proposed CDMAN method.

III. Experiments

In this section, we first describe the three image–text datasets and evaluation metrics. Then, we introduce the experimental details necessary to implement the training process. After that, numerous parametric and ablation experiments are conducted to verify the feasibility of the proposed module. Finally, we compare and analyze the proposed CDMAN method with existing MCR techniques to demonstrate the effectiveness of our approach.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:53:06 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Proposed CDMAN Algorithm

Input:

N pairwise units $\mathcal{P} = \{I_i, T_i\}_{n=1}^{N}$;

Learning rate lr, tradeoff coefficients ω and ρ, iterative epoch K.

Output:

The parameters θ_v and θ_t in the feature extraction step;

The parameters Θ_v and Θ_t in the global-local alignment learning step.

Initialization:

The parameter θ_v is initialized by the pretrained PVT network;

The parameter θ_t is initialized by the pretrained BERT network;

The parameters Θ_v and Θ_t are randomly initialized by glorot_uniform distribution.

Repeat:

1: Utilize cross-modal pairwise construction to select pairwise units $\{I_i, T_i\}$;

2: Compute $F_l(I_n)$, $F_g(I_n)$, $F_l(T_n)$ and $F_g(T_n)$ by feature extraction;

3: Learn fine-grained cues injection to get the fine-grained similarity score S_{TI} and S_{IT} according to Eq. (11) and Eq. (16);

4: Learn coarse-grained association to get the coarse-grained similarity score S_g via Eq. (18);

5: Compute the multigranularity cooperative similarity score $S(I, T)$ according to Eq. (17);

6: Calculate the objective function $L(I, T)$ by using Eq. (19);

7: Update the parameters θ_v, θ_t, Θ_v and Θ_t by utilizing Adam optimizer.

Until:

A fixed iterative epoch K or a stopping criterion is satisfied.

Return: θ_v, θ_t, Θ_v and Θ_t

A. Datasets and Evaluation Metrics

To conduct a comprehensive analysis, we performed a series of extensive experiments on three mainstream RS datasets, namely the RSICD image–text dataset [39], the RSITMD image–text dataset [7], and the UCM image–text dataset [40], to verify the effectiveness of the proposed CDMAN method. Each sample pair in these datasets contains one RS image and five text descriptions. The RSICD image–text dataset comprises over 1000 RS images collected from sources such as Google Earth, Baidu Map, Mapabc, and Tianditu. All images are standardized to a size of 224 × 224 pixels with diverse resolutions. The dataset contains a total of 10 921 RS images. The RSITMD image–text dataset is a more fine-grained and challenging dataset for RS image retrieval. The images were selected from the RSICD dataset, with the rest sourced from Google Earth, containing 4743 images and 23 715 text descriptions. The UCM image–text dataset is one of the earliest datasets proposed by the Chinese Academy of Sciences for RS image retrieval. It contains 2100 RS images, each with dimensions of 256 × 256 pixels, representing 21 scenes sourced from the USGS National Map.

TABLE I

CROSS-MODAL RS RETRIEVAL RESULTS WITH DIFFERENT VALUE OF ω ON THE RSITMD IMAGE–TEXT DATASET

ω	I→T			T→I			mr
	R@1	R@5	R@10	R@1	R@5	R@10	
0.1	15.93	34.96	48.23	12.12	42.30	62.70	36.04
0.2	15.13	36.73	50.09	12.62	42.03	59.49	36.01
0.3	16.00	36.21	49.48	13.23	42.99	60.12	36.34
0.4	15.04	36.28	51.28	12.98	41.82	59.19	36.10
0.5	16.15	37.17	51.33	12.52	41.33	59.56	36.34
0.6	14.82	35.75	50.31	13.24	41.22	58.90	35.71
0.7	15.88	37.04	51.06	12.81	43.04	59.89	36.62
0.8	**16.46**	**37.74**	**51.24**	**13.82**	**41.96**	**59.74**	**36.83**
0.9	16.59	37.61	50.22	12.43	41.37	60.18	36.40
1.0	15.04	37.26	51.19	12.95	41.61	59.37	36.24

To evaluate the performance of our experiments, we employed the topK (R@K) recall metric for both image-as-query text retrieval and text-as-query image retrieval scenarios. This metric allowed us to measure the accuracy of the retrieval system by examining the percentage of correct matches within the top-K results. The topK (R@K) is defined as the K samples closest to the queried ground truth. In this experiment, K values of 1, 5, and 10 were selected for calculation. Additionally, we leverage the mean of the six recall rates of R@K as mr, which helps evaluate the overall performance of the model.

B. Experimental Details

To conduct our experiments, we implement the proposed CDMAN approach using the PyTorch library, an open-source deep learning framework. All experiments are conducted on a workstation equipped with an NVIDIA GeForce RTX 3090 GPU. Through a series of data preprocessing operations, the input RS image size is adjusted to 256 × 256. The embedding dimension for text is set to 256. The BERT pretraining model [41] is used for feature extraction to obtain text features. The global feature dimension for a single modality is 512, and the local feature dimension is also 512. The total similarity between RS images and text is influenced by the weighted parameters ω and ρ. And the weighted parameters ω and ρ are set as 0.8 and 1, respectively. The validity of these parameters is verified through parameter experiments. During the training process, we use the Adam optimizer to train the network. The training process consists of 20 epochs. To effectively optimize the network, we implement a learning rate decay strategy. Specifically, the initial learning rate is set to 1e-4 and decreases by a factor of 0.5 every five epochs. This decay strategy enhances the learning process and helps achieve better results.

C. Parameter Analysis

To obtain optimal weighted parameters ω and ρ in (17), parameter experiments are conducted on the RSITMD

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:53:06 UTC from IEEE Xplore. Restrictions apply.

TABLE II
CROSS-MODAL RS RETRIEVAL RESULTS WITH DIFFERENT VALUE OF ρ ON THE RSITMD IMAGE–TEXT DATASET

ρ	I→T			T→I			mr
	R@1	R@5	R@10	R@1	R@5	R@10	
0.1	14.87	35.27	49.56	12.78	41.88	58.79	35.52
0.2	14.91	36.02	50.00	12.53	41.63	61.50	36.09
0.3	15.04	34.73	50.15	12.67	42.14	59.57	35.72
0.4	14.97	35.47	47.64	13.02	42.37	59.25	35.45
0.5	15.71	36.36	51.84	13.82	42.30	59.60	36.61
0.6	14.60	36.43	51.11	12.85	42.83	61.02	36.47
0.7	14.82	37.39	50.37	12.54	42.40	60.41	36.32
0.8	15.93	36.06	50.96	13.50	42.55	60.32	36.55
0.9	14.56	36.90	50.31	12.87	42.73	60.73	36.35
1.0	**16.64**	**37.74**	**51.24**	**13.82**	**41.96**	**59.74**	**36.83**

image–text dataset. R@K and mr are applied as metrics to choose the optimal parameters in parameter analysis. First, the parameter ρ is set to a fixed value of 1. Table I presents the metrics for varying values of the parameter ω on the RSITMD image–text dataset. According to Table I, the proposed method achieves the highest mr when ω is set to 0.8. Next, ω is set to 0.8, and the metrics for different values of ρ are obtained on the RSITMD image–text dataset, as shown in Table II. Based on the information in Table I, it is evident that the CDMAN method achieves the optimal mr when ρ is set to 1. As a result, the proposed CDMAN method utilizes parameter values of 0.8 for ω and 1 for ρ.

D. Evaluation of Different Factors

To evaluate the effectiveness of the fine-grained cues injection and multigranularity cooperative association terms in the proposed CDMAN method, we conduct the following experiments: First, we utilize our proposed method to learn the fit objective function for remotely sensed image and textual modality, excluding the fine-grained cues injection term and the multigranularity cooperative association term (i.e., CDMAN-A). Second, we exploit the method in this article to perform feature matching learning using only the textual cues injection term without considering the fine-grained bidirectional cues injection term (i.e., CDMAN-B). Third, we apply the method proposed in this article to cross-modal matching learning using only the visual cues injection term without considering the fine-grained bidirectional cues injection term (i.e., CDMAN-C). Finally, we carry out all parts of the approach proposed in this article (i.e., CDMAN). Table III presents the comparative results of CDMAN-A, CDMAN-B, CDMAN-C, and CDMAN on the RSITMD dataset. "I→T" denotes queries in the imaging modality with a text database, while "T→I" represents queries in the text modality with an image database.

The comparison results in Table III indicate that the CDMAN method we propose achieves better performance than CDMAN-A, CDMAN-B, and CDMAN-C. For example, in retrieving text data using image data, CDMAN can enhance

TABLE III
CROSS-MODAL RS RETRIEVAL RESULTS WITH DIFFERENT ALIGNMENT WAY ON THE RSITMD IMAGE–TEXT DATASET

Alignment	I→T			T→I			mr
	R@1	R@5	R@10	R@1	R@5	R@10	
CDMAN-A	15.18	30.06	50.80	13.20	42.35	59.64	36.20
CDMAN-B	15.66	36.64	50.00	13.33	42.40	60.19	36.37
CDMAN-C	15.04	36.28	51.06	13.42	42.68	59.65	36.36
CDMAN	16.46	37.74	51.24	13.82	41.96	59.74	36.83

R@1 to 16.46% from 15.18% achieved by CDMAN-A, 15.66% implemented by CDMAN-B, and 15.04% implemented by CDMAN-C. In leveraging text data to retrieve image data, CDMAN can enhance R@1 to 13.82% from 13.20% carried out by CDMAN-A, 13.33% implemented by CDMAN-B, and 13.42% implemented by CDMAN-C. The injection of cues enables the full alignment of positive samples between modalities while increasing the learning difficulty of negative samples. Therefore, our method can better align the features between RS images and text, thereby improving retrieval accuracy. Additionally, the multigranularity association process neither misses the association of details nor compromises global semantic consistency. This design enables our CDMAN method to excel in MCR tasks and achieve higher performance than the other variant methods.

E. Methods Comparison

To assess the effectiveness of our CDMAN method, we conducted a comparative analysis with several state-of-the-art approaches. The introduction to these methods is as follows.

1) *VSE++ [42]:* VSE++ utilizes VGG19 for extracting image features and a bidirectional GRU for extracting text features. The model is trained using contrastive loss.
2) *SCAN [43]:* SCAN performs local feature extraction using a bottom-up attention mechanism by incorporating object detection. In this experiment, we compared CDMAN with the SCAN t2i and SCAN i2t approaches described in the mentioned article. In image processing, Faster R-CNN serves as the underlying framework for generating candidate regions, while ResNet is employed for feature extraction. For text processing, SCAN utilizes a Bi-GRU as the foundational architecture for extracting text features.
3) *CAMP [44]:* CAMP identifies key points in images and text, enabling interaction to better capture matching image–text pairs for MCR. In this experiment, we compared CDMAN with the CAMP-bce and CAMP-triplet approaches described in this article. The backbone of this method for image and text processing is consistent with that of the SCAN method.
4) *MTFN [45]:* MTFN utilizes cross-modal embedding to perform tensor infusion to enhance the modeling of different modalities. Moreover, a reranking algorithm is employed to improve the efficiency of the model. This method uses Faster R-CNN and ResNet as the backbone

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:53:06 UTC from IEEE Xplore. Restrictions apply.

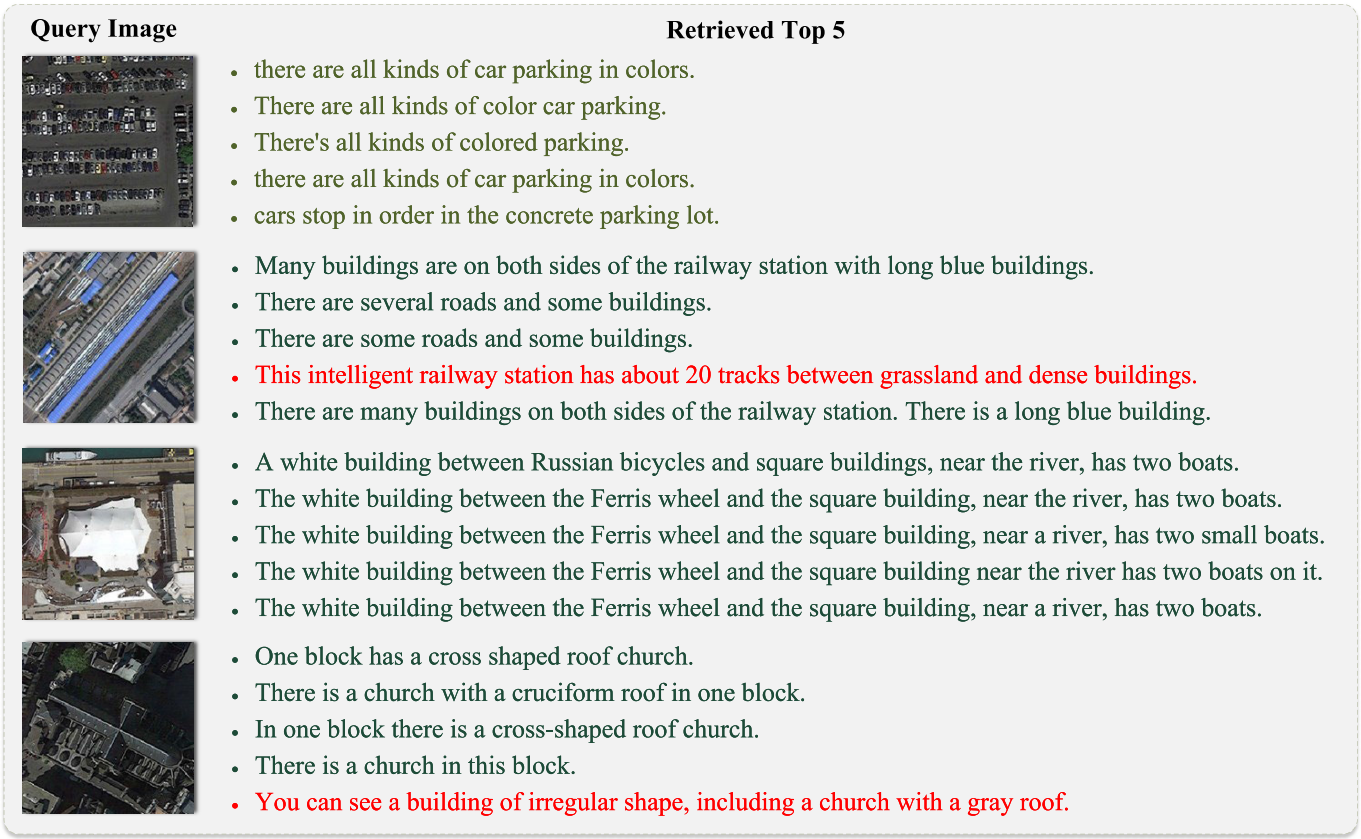


Fig. 5. Top-5 retrieval texts with the given query images on the four datasets. The right retrieval texts are annotated with green. The wrong retrieval texts are annotated with red.

TABLE IV

EXPERIMENTAL RESULTS AND METHODS' BACKBONE BETWEEN CDMAN AND COMPARED STATE-OF-THE-ARTS ON THE RSICD IMAGE–TEXT DATASET AND THE RSITMD IMAGE–TEXT DATASET

Method	Backbone		RSICD dataset							RSITMD dataset						
	Image	Text	I→T			T→I			mr	I→T			T→I			mr
			R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
VSE++	ResNet152	VGG19	3.38	9.51	17.46	2.82	11.32	18.10	10.43	10.38	27.65	39.60	7.79	24.87	38.67	24.83
SCAN t2i	Faster R-CNN, ResNet	Bi-GRU	4.39	10.90	17.64	3.91	16.20	26.49	13.25	10.18	28.53	38.49	10.10	28.98	43.53	26.64
SCAN i2t			5.85	12.89	19.84	3.71	16.4	26.73	14.23	11.06	25.88	39.38	9.82	29.38	42.12	26.82
CAMP-triple	Faster R-CNN, ResNet	Bi-GRU	5.12	12.89	21.12	4.15	15.23	27.81	14.39	11.73	26.99	38.05	8.27	27.79	44.34	26.20
CAMP-bce			4.2	10.24	15.45	2.72	12.76	22.89	11.38	9.07	23.01	33.19	5.22	23.32	38.36	22.03
MTFN	Faster R-CNN, ResNet	GRU	5.02	12.52	19.74	4.9	17.17	29.49	14.81	10.40	27.65	36.28	9.96	31.37	45.84	26.92
CLIP	ResNet50	Transformer	7.96	21.41	32.48	5.42	20.02	33.19	20.08	14.89	33.55	45.64	9.81	32.68	47.43	30.67
LW-MCR	SqueezeNet	RNNS	4.39	13.35	20.29	4.3	18.85	32.34	15.59	9.73	26.77	37.61	9.25	34.07	54.03	28.58
AMFMN	ResNet18	Bi-GRU	5.39	15.08	23.4	4.9	18.28	31.44	16.42	10.63	24.78	41.81	11.51	34.09	54.87	29.72
SWAN	ResNet50	Bi-GRU	7.41	20.13	30.86	5.56	22.26	37.41	20.61	13.35	32.15	46.9	11.24	40.40	60.60	34.11
MGRM-EL	ResNet18	Skipthought	7.41	23.24	35.32	5.75	21.23	35.55	21.42	13.51	31.87	46.27	11.11	37.22	56.61	32.76
TGKT	CLIP-ViT	CLIP-Transformer	8.14	23.06	36.51	6.81	24.52	39.30	23.06	15.93	35.62	50.88	12.08	42.26	62.21	36.50
VGSGN	ResNet50	Bi-GRU	8.33	21.87	32.57	6.53	23.13	36.85	21.55	14.16	34.96	50.66	13.23	42.57	63.41	36.50
DOVE-S	ResNet18	Bi-GRU	7.62	19.85	30.65	5.84	21.87	36.54	20.40	15.71	31.86	44.69	12.09	42.39	59.16	34.32
DOVE	ResNet50		8.66	22.35	34.95	6.04	23.95	40.35	21.72	**16.81**	36.80	49.93	12.20	**44.13**	**66.50**	**37.73**
CDMAN	PVT	Bert	**10.23**	**24.94**	**36.83**	**7.13**	**25.38**	**40.35**	**24.14**	16.46	**37.74**	**51.24**	**13.82**	41.96	59.74	36.83

for image feature extraction. For text processing, MTFN uses GRU as the backbone for extracting text features.

5) *AMFMN [7]:* AMFMN utilizes a multiscale and redundant information filtering structure to process high-level image features, positioning image and text information through semantic alignment. For image processing, ResNet18 with a shallow structure is used as the backbone for image feature extraction. For text processing, AMFMN uses GRU as the backbone for extracting text features.

6) *LW-MCR [15]:* LW-MCR proposes an LW multiscale retrieval algorithm to extract dark knowledge from different layers. In image processing, SqueezeNet is used as the backbone for image feature extraction, making it suitable for scenarios with low computing resource requirements. For text processing, LW-MCR uses RNNs as the backbone for extracting text features.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:53:06 UTC from IEEE Xplore. Restrictions apply.

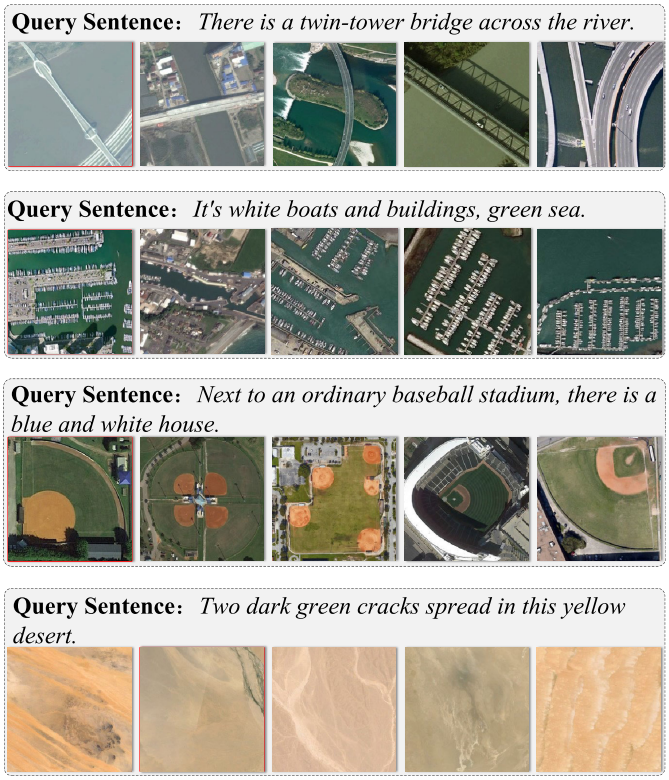


Fig. 6. Top-5 retrieval images with the given some query texts on the four datasets. The images are ranked from left to right, with ranks 1–5. The image within the red box corresponds to the ground truth.

7) *CLIP [46]:* CLIP is a pretrained model designed for cross-modal learning, specifically pairing images and text using contrastive learning. We use CLIP with a pretrained ResNet50 and Transformer as the backbone to perform RS image–text retrieval tasks.
8) *SWAN [47]:* The SWAN framework is designed with a visual multiscale fusion module, a scene fine-grained sensing module, and a textual coarse-grained enhancement module, effectively reducing semantic confusion in RS images and enhancing MCR performance. This method uses ResNet50 as the backbone for image feature extraction and Bi-GRU as the backbone for text feature extraction.
9) *MGRM-EL [48]:* By employing ResNet18 as the backbone for image feature extraction and utilizing Skipthought as the backbone for text feature extraction, MGRM-EL leverages mask-guided relation modeling with entity loss. This is coupled with a Transformer architecture and a mask-guided attention strategy to enhance unimodal learning capabilities while addressing information redundancy in the retrieval process.
10) *TGKT [49]:* TGKT, a text-guided knowledge transfer method, uses CLIP as the backbone for images and text, transfers semantic knowledge from natural to RS domains, and bridges the semantic gap with textual information to enhance image features.
11) *VGSGN [50]:* The Visual Global-Salient Guided Network aligns text and image features by using global and salient image features to guide text features, enhancing RS text–image retrieval performance.

TABLE V
EXPERIMENTAL RESULTS BETWEEN CDMAN AND COMPARED STATE-OF-THE-ARTS ON THE UCM IMAGE–TEXT DATASET

Method	UCM dataset						mr
	I→T			T→I			
	R@1	R@5	R@10	R@1	R@5	R@10	
VSE++	12.38	44.76	65.71	10.10	31.80	56.85	36.93
SCAN t2i	14.29	45.71	67.62	12.76	50.38	77.24	44.67
SCAN i2t	12.85	47.14	69.52	12.48	46.86	71.71	43.43
CAMP-triplet	10.95	44.29	65.71	9.90	46.19	76.29	42.22
CAMP-bce	14.76	46.19	67.62	11.71	47.24	76.00	43.92
MTFN	10.47	**47.62**	64.29	**14.19**	52.38	78.95	44.65
CLIP	13.81	47.14	62.38	9.90	46.19	73.14	42.10
LW-MCR	**18.10**	47.14	63.81	13.14	50.38	79.52	45.35
AMFMN	16.67	45.71	68.57	12.86	**53.24**	79.43	46.08
SWAN	-	-	-	-	-	-	-
MGRM-EL	-	-	-	-	-	-	-
TGKT	-	-	-	-	-	-	-
VGSGN	-	-	-	-	-	-	-
DOVE-S	-	-	-	-	-	-	-
DOVE	-	-	-	-	-	-	-
CDMAN	10.95	47.43	**73.53**	12.23	51.83	**85.14**	**46.90**

* To ensure transparency and reproducibility, we only include studies with publicly available source code for comparison. *

12) *DOVE [51]:* The DOVE model improves RS image–text retrieval by aligning visual and textual representations in a redundancy-free latent space using regional attention and an LW text assistant. DOVE-S is a small-size DOVE model, using ResNet-18 as the image encoder.

Tables IV and V present the comparative experimental results by using texts to retrieve images and the comparative experimental results by using images to retrieve texts on the RSICD dataset, the RSITMD dataset, and the UCM dataset, respectively. Figs. 5 and 6 show the top five retrieval results of the proposed CDMAN approach on the RSITMD dataset.

We can obviously take note of the following.

1) Compared with other MCR algorithms, our proposed CDMAN algorithm demonstrates superior performance across various metrics. Specifically, on the RSICD and UCM datasets, it achieves the highest retrieval accuracy in terms of mean average precision. The RSITMD dataset achieves the second-best retrieval accuracy in terms of mean precision.
2) Current cross-modal search methods based on multimodal pretrained models can achieve better search performance compared to traditional cross-modal search methods utilizing the two-tower model. For instance, when retrieving text data using image data, CLIP has improved the recall rate (mr) of methods such as VSE++, SCAN, CAMP, and MTFN on the RSICD and RSITMD datasets to 20.08% and 30.64%, respectively. Meanwhile, on small-scale datasets like UCM, the CLIP model does not surpass traditional two-tower models. However, because CLIP is not specifically tailored to handle the unique characteristics of RS data, its overall performance falls short of that of RS-specific retrieval

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:53:06 UTC from IEEE Xplore. Restrictions apply.

models such as SWAN, MGRM-EL, TGKT, VGSGN, and DOVE.

3) Most current CMRMs applied to RS utilize the two-tower model, and the CDMAN method proposed in this article follows suit. In the natural image domain, multimodal pretrained models outperform the two-tower model. However, as seen from the table, compared to CLIP, the CDMAN method proposed in this article increases the mr from 20.08% to 24.14% on the RSICD dataset, from 30.64% to 36.83% on the RSITMD dataset, and from 42.10% to 46.90% on the UCM dataset. Additionally, compared to the LW-MCR, AMFMN, SWAN, MGRM-EL, TGKT, VGSGN, and DOVE-S methods specifically designed for the RS domain, the CDMAN method improves the mr on the RSICD dataset by 8.55%, 7.72%, 3.53%, 2.72%, 1.08%, 2.59%, and 3.74%, respectively, and on the RSITMD dataset by 8.25%, 7.11%, 2.72%, 4.07%, 0.33%, 0.33% and 2.51%, respectively. It can be observed that our method shows improvements over LW-MCR and AMFMN on the UCM dataset. Other methods did not release their code and checkpoints, so we were unable to compare them on the UCM dataset.
4) Furthermore, we are analyzing the strengths and weaknesses of the existing method backbone. VSE++ employs ResNet152 and VGG19 as its image and text backbone networks. This older model tends to show lower performance, particularly in the R@1 metric, indicating less accuracy in retrieval tasks. The SCAN and CAMP methods utilize a combination of Faster R-CNN and ResNet with Bi-GRU as their backbone networks, highlighting the effectiveness of integrating object detection networks with deep feature networks for MCR tasks. CLIP leverages ResNet50 along with Transformer models, demonstrating superior performance, especially on the RSITMD dataset. This likely stems from the Transformer's proficiency in processing sequential data, making it well-suited for complex text descriptions. LW-MCR and AMFMN opt for more LW image networks such as SqueezeNet and ResNet18. While these models are computationally efficient, they often do not match the performance of deeper networks. MGRM-EL and TKGT employ more recent architectures like CLIP-ViT and CLIP-Transformer, which deliver high performance, illustrating the advantages of modern visual and language models in these tasks. Our CDMAN framework employs a dual-Transformer architecture incorporating PVT and Bert, which has shown remarkable performance, particularly on the RSTIMD dataset. Experimental results indicate that the integration of PVT with Bert's robust text processing capabilities leads to more refined feature representation, contributing to the system's superior performance.

Our model notably achieves significant improvements in precision rate R@1, likely due to the CDMAN method's ability to effectively learn and distinguish positive from negative samples through its hard sample construction module. Comparative experiments show that the proposed multigranularity association strategy in CDMAN outperforms single-association methods across various metrics. However, CDMAN shows a slight decline in text retrieval accuracy on the RSITMD dataset, likely due to the simpler information carried by text and the challenge of matching it with complex images. Additionally, the dataset's structure—where one text corresponds to a single image, but an image corresponds to five sentences—may contribute to this issue. While complex models perform well on large-scale datasets, their potential is often constrained by data limitations, affecting their expressive capacity. Small datasets may fail to capture the true data distribution, leading to inaccuracies in retrieving RS data with high intraclass variability and low interclass variability. Consequently, the small-scale UCM dataset poses greater challenges compared to larger datasets like RSITMD and RSICD. Despite this, our method excels on the UCM dataset, demonstrating its ability to effectively learn distinctions between positive and negative samples.

In future work, we aim to improve text retrieval recall by focusing on the dataset, including creating balanced text–image pairings and leveraging text modality more effectively, especially for complex images. Our goal is to develop robust, accurate MCR methods to address the challenges of RS data.

IV. Conclusion

In this article, we propose an innovative MCR method for RS images and texts, leveraging modal cues to enhance positive sample features, mitigate negative sample features, and achieve multigranularity collaborative association between RS images and texts. First, the CDMAN method introduces a novel approach for aligning positive samples and enhancing difficult samples based on fine-grained modal cues, aiming to achieve precise alignment of positive samples and the construction of challenging samples. Second, multigranularity association learning is implemented to establish global and detailed associations between the extracted image and text features. This approach enables detailed-level associations while maintaining overall semantic consistency in global associations, thus mitigating the impact of redundant details.

While our method improves RS text–image retrieval by incorporating multigranularity alignment, association, and effective negative sample learning, it has limitations. First, we focus solely on explicit cross-modal alignment, overlooking the complementarity between modalities. Cross-modal relationships should go beyond object-level alignment to include the cooperative nature of modality information, a challenging area requiring further research. Second, we have not utilized multimodal pretrained models like CLIP, which excel in cross-modal tasks by capturing diverse visual-language relationships. Integrating such models could enhance the initial representation of RS text and images in our framework.

References

[1] L. Zhang, Y. Zhang, R. Hong, and Q. Tian, "Full-space local topology extraction for cross-modal retrieval," *IEEE Trans. Image Process.*, vol. 24, no. 7, pp. 2212–2224, Jul. 2015.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:53:06 UTC from IEEE Xplore. Restrictions apply.

[2] X. Lu, Y. Chen, and X. Li, "Siamese dilated inception hashing with intra-group correlation enhancement for image retrieval," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 8, pp. 3032–3046, Aug. 2020.

[3] M. Guo, C. Zhou, and J. Liu, "Jointly learning of visual and auditory: A new approach for RS image and audio cross-modal retrieval," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 11, pp. 4644–4654, Nov. 2019.

[4] L. Xie, P. Pan, and Y. Lu, "Analyzing semantic correlation for cross-modal retrieval," *Multimedia Syst.*, vol. 21, no. 6, pp. 525–539, Nov. 2015.

[5] K. Li, G.-J. Qi, J. Ye, and K. A. Hua, "Linear subspace ranking hashing for cross-modal retrieval," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 9, pp. 1825–1838, Sep. 2017.

[6] Y. Chen, X. Lu, and Y. Feng, "Deep voice-visual cross-modal retrieval with deep feature similarity learning," in *Proc. Chin. Conf. Pattern Recognit. Comput. Vis. (PRCV)*, 2019, pp. 454–465.

[7] Z. Yuan et al., "Exploring a fine-grained multiscale method for cross-modal remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4404119.

[8] Z. Yuan et al., "Remote sensing cross-modal text-image retrieval based on global and local information," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620616.

[9] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.

[10] D. F. Gordon and M. Desjardins, "Evaluation and selection of biases in machine learning," *Mach. Learn.*, vol. 20, nos. 1–2, pp. 5–22, 1995, doi: 10.1007/BF00993472.

[11] X. Tang, Y. Wang, J. Ma, X. Zhang, F. Liu, and L. Jiao, "Interacting-enhancing feature transformer for cross-modal remote-sensing image and text retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5611715.

[12] M. M. A. Rahhal, M. A. Bencherif, Y. Bazi, A. Alharbi, and M. L. Mekhalfi, "Contrasting dual transformer architectures for multi-modal remote sensing image retrieval," *Appl. Sci.*, vol. 13, no. 1, p. 282, Dec. 2022.

[13] M. M. A. Rahhal, Y. Bazi, N. A. Alsharif, L. Bashmal, N. Alajlan, and F. Melgani, "Multilanguage transformer for improved text to remote sensing image retrieval," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9115–9126, 2022.

[14] Y. Wang, J. Ma, M. Li, X. Tang, X. Han, and L. Jiao, "Multi-scale interactive transformer for remote sensing cross-modal image-text retrieval," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 839–842.

[15] Z. Yuan et al., "A lightweight multi-scale crossmodal text-image retrieval method in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5612819, doi: 10.1109/TGRS.2021.3124252.

[16] Q. Cheng, Y. Zhou, P. Fu, Y. Xu, and L. Zhang, "A deep semantic alignment network for the cross-modal image-text retrieval in remote sensing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4284–4297, 2021, doi: 10.1109/JSTARS.2021.3070872.

[17] Y. Lv, W. Xiong, X. Zhang, and Y. Cui, "Fusion-based correlation learning model for cross-modal remote sensing image retrieval," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3131592.

[18] J. Gu, J. Cai, S. Joty, L. Niu, and G. Wang, "Look, imagine and match: Improving textual-visual cross-modal retrieval with generative models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7181–7189.

[19] Q. Jiang and W. Li, "Deep cross-modal hashing," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3270–3278.

[20] W. Cao, Q. Lin, Z. He, and Z. He, "Hybrid representation learning for cross-modal retrieval," *Neurocomputing*, vol. 345, pp. 45–57, Jun. 2019.

[21] Y. Cui, Y. Xu, R. Peng, and D. Wu, "Layer normalization for TSK fuzzy system optimization in regression problems," *IEEE Trans. Fuzzy Syst.*, vol. 31, no. 1, pp. 254–264, Jan. 2023, doi: 10.1109/TFUZZ.2022.3185464.

[22] J. Xu, X. Sun, Z. Zhang, G. Zhao, and J. Lin, "Understanding and improving layer normalization," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, Vancouver, BC, Canada, H. M. Wallach, H. Larochelle, A. Beygelzimer, F. D'Alché-Buc, E. B. Fox, and R. Garnett, Eds., Dec. 2019, pp. 4383–4393. [Online]. Available: https://proceedings.neurips.cc/paper/2019/hash/2f4fe03d77724a7217006e5d16728874-Abstract.html

[23] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–12.

[24] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Hum. Lang. Technol.*, J. Burstein, C. Doran, and T. Solorio, Eds., Stroudsburg, PA, USA: Association for Computational Linguistics, 2019, pp. 4171–4186, doi: 10.18653/v1/n19-1423.

[25] X. Liu, Z. Hu, H. Ling, and Y.-M. Cheung, "MTFH: A matrix tri-factorization hashing framework for efficient cross-modal retrieval," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 3, pp. 964–981, Mar. 2021.

[26] X. Xu, F. Shen, Y. Yang, H. T. Shen, and X. Li, "Learning discriminative binary codes for large-scale cross-modal retrieval," *IEEE Trans. Image Process.*, vol. 26, no. 5, pp. 2494–2507, May 2017.

[27] L. Wu, Y. Wang, and L. Shao, "Cycle-consistent deep generative hashing for cross-modal retrieval," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1602–1612, Apr. 2019.

[28] A. Karpathy, A. Joulin, and L. Fei-Fei, "Deep fragment embeddings for bidirectional image sentence mapping," in *Proc. Adv. Neural Inf. Process. Syst.*, Z. Ghahramani, M. Welling, C. Cortes, N. D. Lawrence, and K. Q. Weinberger, Eds., 2014, pp. 1889–1897. [Online]. Available: https://proceedings.neurips.cc/paper/2014/hash/84d2004bf28a2095230e8e14993d398d-Abstract.html

[29] T. Luong, H. Pham, and C. D. Manning, "Effective approaches to attention-based neural machine translation," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, L. Màrquez, C. Callison-Burch, J. Su, D. Pighin, and Y. Marton, Eds., Stroudsburg, PA, USA: Association for Computational Linguistics, 2015, pp. 1412–1421, doi: 10.18653/v1/d15-1166.

[30] J. Chorowski, D. Bahdanau, D. Serdyuk, K. Cho, and Y. Bengio, "Attention-based models for speech recognition," in *Proc. 28th Int. Conf. Neural Inf. Process. Syst.*, C. Cortes, N. D. Lawrence, D. D. Lee, M. Sugiyama, and R. Garnett, Eds., 2015, pp. 577–585. [Online]. Available: https://proceedings.neurips.cc/paper/2015/hash/1068c6e4c8051cfd4e9ea8072e3189e2-Abstract.html

[31] B.-H. Juang, W. Hou, and C.-H. Lee, "Minimum classification error rate methods for speech recognition," *IEEE Trans. Speech Audio Process.*, vol. 5, no. 3, pp. 257–265, May 1997, doi: 10.1109/89.568732.

[32] X. He, L. Deng, and W. Chou, "Discriminative learning in sequential pattern recognition," *IEEE Signal Process. Mag.*, vol. 25, no. 5, pp. 14–36, Sep. 2008, doi: 10.1109/MSP.2008.926652.

[33] A. Karpathy and L. Fei-Fei, "Deep visual-semantic alignments for generating image descriptions," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 664–676, Apr. 2017, doi: 10.1109/TPAMI.2016.2598339.

[34] Y. Chen, J. Huang, S. Xiong, and X. Lu, "Fine aligned discriminative hashing for remote sensing image-audio retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4702012, doi: 10.1109/TGRS.2023.3269300.

[35] D. Xie, C. Deng, C. Li, X. Liu, and D. Tao, "Multi-task consistency-preserving adversarial hashing for cross-modal retrieval," *IEEE Trans. Image Process.*, vol. 29, pp. 3626–3637, 2020.

[36] M. Mao, J. Lu, G. Zhang, and J. Zhang, "Multirelational social recommendations via multigraph ranking," *IEEE Trans. Cybern.*, vol. 47, no. 12, pp. 4049–4061, Dec. 2017.

[37] H. Liu, R. Wang, S. Shan, and X. Chen, "Deep supervised hashing for fast image retrieval," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2064–2072.

[38] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[39] X. Lu, B. Wang, X. Zheng, and X. Li, "Exploring models and data for remote sensing image caption generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2183–2195, Apr. 2018, doi: 10.1109/TGRS.2017.2776321.

[40] B. Qu, X. Li, D. Tao, and X. Lu, "Deep semantic understanding of high resolution remote sensing image," in *Proc. Int. Conf. Comput., Inf. Telecommun. Syst. (CITS)*, Jul. 2016, pp. 1–5, doi: 10.1109/CITS.2016.7546397.

[41] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," *arXiv:1810.04805*, 2018.

[42] F. Faghri, D. J. Fleet, J. R. Kiros, and S. Fidler, "VSE++: Improving visual-semantic embeddings with hard negatives," 2017, *arXiv:1707.05612*.

[43] K.-H. Lee, X. Chen, G. Hua, H. Hu, and X. He, "Stacked cross attention for image-text matching," in *Proc. Eur. Conf. Comp. Vis. (ECCV)*, 2018, pp. 201–216.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:53:06 UTC from IEEE Xplore. Restrictions apply.

[44] Z. Wang et al., "CAMP: Cross-modal adaptive message passing for text-image retrieval," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5764–5773.
[45] T. Wang, X. Xu, Y. Yang, A. Hanjalic, H. T. Shen, and J. Song, "Matching images and text with multi-modal tensor fusion and re-ranking," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 12–20.
[46] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn. (ICML)*, in Proceedings of Machine Learning Research, vol. 139, M. Meila and T. Zhang, Eds., Jul. 2021, pp. 8748–8763. [Online]. Available: http://proceedings.mlr.press/v139/radford21a.html
[47] J. Pan, Q. Ma, and C. Bai, "Reducing semantic confusion: Scene-aware aggregation network for remote sensing cross-modal retrieval," in *Proc. ACM Int. Conf. Multimedia Retr. (ICMR)*, New York, NY, USA, 2023, pp. 398–406, doi: 10.1145/3591106.3592236.
[48] S. Zhang, Y. Li, and S. Mei, "Exploring uni-modal feature learning on entities and relations for remote sensing cross-modal text-image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5626317.
[49] A.-A. Liu et al., "Text-guided knowledge transfer for remote sensing image-text retrieval," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[50] Y. He, X. Xu, H. Chen, J. Li, and F. Pu, "Visual global-salient-guided network for remote sensing image-text retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5641814.
[51] Q. Ma, J. Pan, and C. Bai, "Direction-oriented visual–semantic embedding model for remote sensing image–text retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4704014.



Yaxiong Chen is an Associate Professor with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan, China. His research interests include pattern recognition, machine learning, hyperspectral image analysis, and medical imaging.



Jirui Huang is currently pursuing the Ph.D. degree with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan, China.

Her research interests include computer vision, machine learning, and remote sensing image processing, especially cross-modal retrieval and multimodal learning.



Zhaoyang Sun (Graduate Student Member, IEEE) received the B.E. and M.S. degrees from Wuhan University of Technology, Wuhan, China, in 2018 and 2021, respectively, where he is currently pursuing the Ph.D. degree with the School of Computer Science and Artificial Intelligence.

His research interests include pattern recognition, machine learning, image translation, and image synthesis.



Shengwu Xiong received the B.Sc. degree in computational mathematics and the M.Sc. and Ph.D. degrees in computer software and theory from Wuhan University, Wuhan, China, in 1987, 1997, and 2003, respectively.

Currently, he is a Professor with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan. His research interests include intelligent computing, machine learning, and pattern recognition.



Xiaoqiang Lu (Senior Member, IEEE) is currently a Full Professor with the College of Physics and Information Engineering, Fuzhou University, Fuzhou, China. His research interests include intelligent optical sensing, pattern recognition, machine learning, and hyperspectral image analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:53:06 UTC from IEEE Xplore. Restrictions apply.